

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Physics) Examination November 2019

PD202 Thermal Physics

Date: 08/11/2019 Day: Friday Time: 10.00 am To 10.30 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

- Two identical rectangular strips one of copper and the other of steel are riveted together to form bimetallic strips. What will happen on heating these strips?

(i) It will remain straight	(ii) It will get twisted
(iii) It will bend with steel on convex side	(iv) It will bend with copper on convex side
- When a metal-sheet with a circular hole is heated, the hole

(i) Gets larger	(ii) Gets smaller
(iii) Remains the same	(iv) Gets deformed
- A gas behaves more closely as an ideal gas at

(i) low pressure and low temperature	(ii) low pressure and high temperature
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- (iii) high pressure and low temperature (iv) high pressure and high temperature
4. The rms speed oxygen molecule in a gas is v . If the temperature is doubled and the oxygen molecule dissociate into oxygen atoms, the rms speed will become _____.
 (i) v (ii) $\sqrt{2}v$
 (iii) $2v$ (iv) $4v$
5. When a hot liquid is mixed with a cold liquid, the temperature of the mixture
 (i) first decreases then becomes constant (ii) first increases then becomes constant
 (iii) continuously increases (iv) is undefined for sometimes and then becomes nearly constant
6. Two bodies at different temperature s are mixed in a calorimeter. Which of the following quantities remain conserved?
 (i) Sum of the temperatures of the two bodies (ii) Total heat of the two bodies
 (iii) Total internal energy of the two bodies (iv) Internal energy of each body
7. A gas is contained in a metallic cylinder fitted with a piston. The piston is suddenly moved in to compress the gas and is maintained at this position. As time passes the pressure of the gas in the cylinder _____.
 (i) Increases (ii) Decreases
 (iii) Remain constant (iv) Increase or decrease depending on the nature of the gas
8. A system can be taken from the initial state p_1, V_1 to the final state p_2, V_2 by two different methods. Let ΔQ and ΔW represent the heat given to the system and the work done by the system. Which of the following must be the same in both the methods?
 (i) ΔQ (ii) ΔW
 (iii) $\Delta Q + \Delta W$ (iv) $\Delta Q - \Delta W$
9. For a solid with a small expansion coefficient,
 (i) $C_p - C_v = R$ (ii) $C_p = C_v$

(iii) C_p is slightly greater than C_v

(iv) C_p is slightly less than C_v

10. Two samples A and B are initially kept in the same state. The sample A is expanded through an adiabatic process and sample B through an isothermal process. The final volumes of the samples are the same. The final pressures in A and B are p_A and p_B respectively. What of the following condition satisfy for both the samples?

(i) $p_A > p_B$

(ii) $p_A = p_B$

(iii) $p_A < p_B$

(iv) none of the above

11. The thermal conductivity of rod depends on

(i) Length

(ii) Mass

(iii) Area of cross section

(iv) Material of the rod

12. Newton's Law of cooling is a special case of

(i) Wien's displacement law

(ii) Kirchhoff's law

(iii) Stefans's law

(iv) Planck's law

13. The necessary condition for a reversible process is _____

(i) the process should be too fast

(ii) the process should be slow so that the working substance should remain in thermal and mechanical equilibrium with the surroundings

(iii) the loss of energy should be zero and it should be quasistatic

(iv) the loss of any type of heat energy should be zero in the whole cycle

14. "Heat cannot flow by itself from a body at lower temperature to a body at higher temperature" is a statement or consequence of _____.

(i) 0th law of thermodynamics

(ii) 1st law of thermodynamics

(iii) 2nd law of thermodynamics

(iv) None of the above

15. Which of the following is not a Maxwell equation?

(i) $(\partial T / \partial V) = -(\partial p / \partial S)$

(ii) $(\partial T / \partial p) = -(\partial V / \partial S)$

(iii) $(\partial p / \partial T) = (\partial S / \partial V)$

(iv) $(\partial V / \partial T) = -(\partial S / \partial p)$

16. Which of the following curves meet at triple point?
- (i) fusion curve and vaporization curve (ii) fusion curve and sublimation curve
- (iii) vaporization curve and sublimation curve (iv) fusion curve and vaporization curve and sublimation curve
17. Identify the statement, true or false.
- The liquid phase can exist at absolute zero.
18. Identify the statement, true or false.
- The solid phase can exist at temperatures above the critical temperature.
19. Which of the following potential replaces entropy and volume of the internal energy equation?
- (i) Helmholtz (ii) Gibbs
- (iii) Enthalpy (iv) Grand canonical
20. Identify the statement, true or false.
- The coefficient of performance of a refrigerator can be greater than one.

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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Physics) Examination November 2019

PD202 Thermal Physics

Date: 08/11/2019 Day: Friday Time: 10.30 am To 1.00 pm Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in SEPARATE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

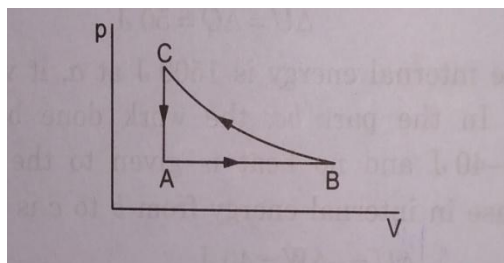
SECTION -I

Q – II

Q – 1 Solve any five problems. (3 marks each)

[15]

- a) A sphere of diameter 7.0 cm and mass 266.5 g floats in a bath of liquid. As the temperature is raised, the sphere begins to sink at a temperature of 35 °C. If the density of the liquid is 1.527 g/cm^3 at 0°C, find the coefficient of cubical expansion of the liquid. Neglect the expansion of the sphere.
- b) 0.040 g of He is kept in a closed container initially at 100 °C. The container is now heated. Neglecting the expansion of the container, calculate the temperature at which the internal energy is increased by 12 J.
- c) The temperature of equal masses of three different liquids A, B and C are 12° C, 19 °C, 28 °C respectively. The temperature A and B are mixed is 16 °C and when B and C are mixed the temperature is 23° C. What will be the temperature when A and C are mixed?
- d) Consider the cyclic process ABCA on a sample of 2.0 mole of an ideal gas as shown in figure. The temperature of the gas at A and B are 300 K and 500 K respectively. A total of 1200 J heat is withdrawn from the sample in the process. Find the work done by the gas in part BC. Take $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$.



- e) An ideal gas at pressure $2.5 \times 10^5 \text{ Pa}$ and temperature 300 K occupies 100 cc. It is adiabatically compressed to half its original volume. Calculate (a) the final pressure, (b) the final temperature and (c) the work done by the gas in the process. Take $\gamma = 1.5$.
- f) One mole of an ideal monatomic gas is kept in a rigid vessel. The vessel is kept inside a steam chamber whose temperature is 97 °C. Initially, the temperature of the gas is 5.0 °C. The walls of the vessel have an inner surface of the area 800 cm^2 and thickness 1.0 cm. If the temperature of the gas increases to 90 °C in 5.0 seconds, find the thermal conductivity

of the material of the walls.

Q – 2 Solve any one problem. (5 marks each)

[5]

- a) A sample of an ideal gas ($\gamma = 1.5$) is compressed adiabatically from a volume of 150 cm^3 to 50 cm^3 . The initial pressure and the initial temperature are 150 kPa and 300 K . Find (a) the number of moles of the gas in the sample, (b) the molar heat capacity at constant volume, (c) the final pressure and temperature, (d) the work done by the gas in the process and (e) the change in internal energy of the gas.
- b) A container of volume 50 cc contains air (mean molecular weight = 28.8 g) and is open to atmosphere where the pressure is 100 kPa . The container is kept in a bath containing melting ice (0°C).
- (a) Find the mass of the air in the container when thermal equilibrium is reached.
- (b) The container is now placed in another bath containing boiling water (100°C). Find the mass of air in the container.
- (c) The container is now closed and placed in the melting-ice bath. Find the pressure of the air when thermal equilibrium is reached.

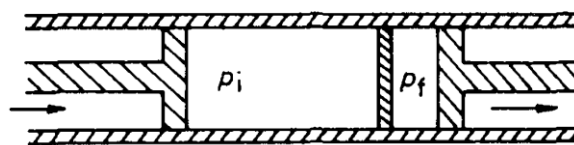
SECTION - II

Q – III Solve the following problems.

[30]

- a) Consider a gas which undergoes an adiabatic expansion (throttling process) from a region of constant pressure p_i and initial volume V_i to a region with constant pressure p_f and final volume V_f (initial volume 0).

[5]



- (a) By considering the work done by the gas in the process, show that the initial and final enthalpies of the gas are equal.
- (b) What can be said about the intermediate states of the system?
- (c) Show for small pressure difference $\Delta p = p_f - p_i$ that the temperature difference between the two regions is given by $\Delta T = \frac{V}{C_p} (T\alpha - 1)\Delta p$, where $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$ and $C_p = \left(\frac{\partial U}{\partial T} \right)_p$.
- (d) Using the above result, discuss the possibilities of using the process to cool either an ideal gas, or a more realistic gas for which $p = RT/(V - b)$. Explain your result.

- b) Determine the ratio $\frac{pv}{RT}$ at the critical point for a gas which obeys the equation of state (Dieterici's equation) $p(V - b) = RT \exp(-a/RTV)$. [4]

Give the numerical answer accurately to two significant figures.

- c) Compute the theoretical minimum amount of mechanical work needed to freeze 1 kilogram of water, if the water and surroundings are initially at a temperature $T^0 = 25^\circ\text{C}$. The surroundings comprise the only large heat reservoir available. (Given: $L_{\text{ice}} = 80 \text{ cal/g}$, $C_p = 1 \text{ cal/g } ^\circ\text{C}$). Derive the vapour pressure equation (Clausius - Clapeyron equation): $\frac{dp}{dT} = ?$ [3]

- d) The following data apply to the triple point of H_2O . [5]

Temperature: 0.01°C , Pressure: 4.6 mmHg

Specific volume of solid: $1.12 \text{ cm}^3/\text{g}$

Specific volume of liquid: $1.00 \text{ cm}^3/\text{g}$

Heat of melting: 80 cal/g

Heat of vaporization: 600 cal/g

- (a) Sketch a p-T diagram for H_2O which need not be to scale but which should be qualitatively correct. Label the various phases and critical points.
- (b) The pressure inside a container enclosing H_2O (which is maintained at $T = -1^\circ\text{C}$) is slowly reduced from an initial value of 105 mmHg . Describe what happens and calculate the pressure at which the phase changes occur. Assume the vapor phase behaves like an ideal gas.
- (c) Calculate the change in specific latent heat with temperature dL/dT at a point (p,T) along a phase equilibrium line. Express your result in terms of L and the specific heat C_p , coefficient of expansion α , and specific volume V of each phase at the original temperature T and pressure p.
- (d) If the specific latent heat at 1 atm pressure on the vaporization curve is 540 cal/g , estimate the change in latent heat 10°C higher than the curve. Assume the vapor can be treated as an ideal gas with rotational degrees of freedom.
- e) (a) By equating the Gibbs free energy or chemical potential on the two sides of the liquid-vapor coexistence curve derive the Clausius – Clapeyron equation : $\frac{dp}{dT} = \frac{q}{T(V_V - V_L)}$, where q is the heat of vaporization per particle and V_L is the volume per particle in the liquid and V_V the volume per particle in the vapor. [5]
- (b) Assuming the vapor follows the ideal gas law and has a density which is much less than that of the liquid, show that $p \sim \exp(-\frac{q}{kt})$, when the heat of vaporization is independent of T.

- f)** (a) Assuming that $1/20$ eV is required to liberate a molecule from the surface of a certain liquid when $T = 300$ K, what is the heat of vaporization in ergs/mole? [3]
 (1 eV = 1.6×10^{-12} erg)
- (b) Estimate the surface tension of a liquid whose heat of vaporization is 10^{10} ergs/g (250 cal/g).
- g)** (a) Using the equation of state $pV = NRT$ and the specific heat per mole $C_v = \frac{3R}{2}$ for a monoatomic ideal gas, find its Helmholtz free energy F as a function of number of moles N , V and T . [5]
- (b) Consider a cylinder separated into two parts by an adiabatic, impermeable piston. Compartments a and b each contains one mole of a monoatomic ideal gas, and their initial volumes are $V_{ai} = 10$ liters and $V_{bi} = 1$ liter, respectively. The cylinder, whose walls allow heat transfer only, is immersed in a large bath at 0°C . The piston is now moved reversibly so that the final volumes are $V_{af} = 6$ and $V_{bf} = 5$ liters. How much work is delivered by (or to) the system?

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